

Lecture notes for March 12 to April 7

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1 Multivariate functions and Its continuity

1.1 Point-set on plane

The familiar point set in $\mathbf{R}$ is interval. The situation is much more complicated in dimension $n \geq 2$.

We've defined distance on plane (or on $\mathbf{R}^3$) before, $d : \mathbf{R}^2 \times \mathbf{R}^2 \rightarrow \mathbf{R}_{\geq 0}$, satisfying

- (Nonnegativity) $\rho(M_1, M_2) \geq 0$, and “=” holds iff $M_1 = M_2$;
- (Symmetric) $\rho(M_1, M_2) = \rho(M_2, M_1)$;
- (Triangle inequality) $\rho(M_1, M_2) \leq \rho(M_1, M_3) + \rho(M_3, M_2)$.

Definition 1.1. The disk $B(M_0, r) = \{M | \rho(M, M_0) < r\}$ is called a neighborhood of M_0 , which is also denoted by $B_r(M_0)$. The set $B_-(M_0, r) = B(M_0, r) \setminus \{M_0\}$ is called a punctured neighborhood of M_0 . We sometimes also use $S(M_0, r) = \{(x, y) | |x - x_0| < r, |y - y_0| < r\}$ as neighborhood.

Both types of neighborhoods can be viewed as the generalization of interval in dimension 2.

Definition 1.2. The set $E \subset \mathbf{R}^2$ is called bounded if $E \subset B(O, R)$ for some $R > 0$, and for bounded set the number

$$\text{diam } E = \sup\{\rho(M', M'') | M', M'' \in E\}$$

is called the diameter of E .

Definition 1.3. Let $E \subset \mathbf{R}^2$ and E^c be its complement in $\mathbf{R}^2$,

- If $\exists r > 0$ such that $B(M, r) \subset E$, then M is called an interior point of E ;
- If $\exists r > 0$ such that $B(M, r) \subset E^c$, then M is called an exterior point of E ;
- If $\forall r > 0$, $B(M, r) \cap E \neq \emptyset$, $B(M, r) \cap E^c \neq \emptyset$, then M is called a boundary point of E ;

The set $\mathring{E} = \{M \in \mathbf{R}^2 | M \text{ is an interior point of } E\}$ is called the interior of E ; and $\partial E = \{M \in \mathbf{R}^2 | M \text{ is a boundary point of } E\}$ is called the boundary of E .

- If $\exists r > 0$ such that $B(M, r) \cap E = \{M\}$, then M is called an isolated point of E
- If $\forall r > 0$, $B_-(M, r) \cap E \neq \emptyset$, then M is called an accumulation point of E ;
- $\overline{E} = E \cup \partial E$ is called the closure of E .

It is easily seen that any interior point must be an accumulation point, and a boundary point is either an isolated point or an accumulation point. On the converse, any accumulation point is either an interior point or an boundary point.

Definition 1.4. Let $\{M_n\}$ be a sequence of points on $\mathbf{R}^2$, and $M_0 \in \mathbf{R}^2$. If

$$\lim_{n \rightarrow \infty} \rho(M_n, M_0) = 0,$$

we call M_n converges to M_0 , or $M_n \rightarrow M_0 (n \rightarrow \infty)$, also denote by $\lim_{n \rightarrow \infty} M_n = M_0$.

Because of the inequality

$$\begin{aligned} |x_n - x_0| &\leq \rho(M_n, M_0), \quad |y_n - y_0| \leq \rho(M_n, M_0) \\ \rho(M_n, M_0) &\leq |x_n - x_0| + |y_n - y_0|, \end{aligned}$$

we conclude that $\lim_{n \rightarrow \infty} M_n = M_0$ iff $\lim_{n \rightarrow \infty} x_n = x_0$ and $\lim_{n \rightarrow \infty} y_n = y_0$, i.e. the point sequence converges iff all of its coordinate components converge.

A sequence of points on $\mathbf{R}$ can only approach one point from left and right direction, however, we should remark that the way of approaching a point on a plane can be more complicated, for instance $M_n = (e^{-n} \cos n, e^{-n} \sin n)$ spirally converges to O . As an analogy to Cauchy sequence for real numbers, we can also define Cauchy sequence of points: if $\forall \varepsilon > 0$, $\exists N = N(\varepsilon) \in \mathbf{N}^+$, s.t. $\forall n, m > N$ we have $\rho(M_n, M_m) < \varepsilon$.

Theorem 1.5.

- (Completeness) All Cauchy sequence converges on $\mathbf{R}^2$;
- (Bolzano-Weierstrass Theorem) Any bounded sequence of points in $\mathbf{R}^2$ has a convergent subsequence.

Definition 1.6. If $E = \mathring{E}$, then E is called an open set. If E^c is open, then E is called a closed set.

Example 1.7. The set $B(M, r)$ (with $r > 0$) and $B_-(M, r)$ are both open; and $B(M, r) \cup \partial B(M, r)$ is closed.

Proposition 1.8. *The following properties and characterizations of open set and closed set hold:*

1. *If E_1, E_2 are open, then $E_1 \cap E_2$ is open; if $\{E_\alpha\}_{\alpha \in \Lambda}$ is a family of open sets, then $\cup_{\alpha \in \Lambda} E_\alpha$ is open;*
2. *If E_1, E_2 are closed, then $E_1 \cup E_2$ is closed; if $\{E_\alpha\}_{\alpha \in \Lambda}$ is a family of closed sets, then $\cap_{\alpha \in \Lambda} E_\alpha$ is closed;*
3. *E is open iff $E \cap \partial E = \emptyset$, i.e. $\partial E \subset E^c$;*
4. *E is closed iff $E^c \cap \partial E = \emptyset$, i.e. $\partial E \subset E$;*
5. *E is closed iff E contains all of its accumulation points.*

Proof. Item 1: For any $M \in E_1 \cap E_2$ there exists $r_1, r_2 > 0$ such that $B(M, r_1) \subset E_1$ and $B(M, r_2) \subset E_2$, therefore $B(M, r) \subset E_1 \cap E_2$ for $r = \min\{r_1, r_2\}$. For another statement, let $M \in \cup_{\alpha \in \Lambda} E_\alpha$, then $M \in E_\alpha$ for some $\alpha \in \Lambda$, thus there exists $r_\alpha > 0$ such that $B(M, r_\alpha) \subset E_\alpha$ by the openness of E_α . This implies that $B(M, r_\alpha) \subset \cup_{\alpha \in \Lambda} E_\alpha$.

Item 2: similar to item 1.

Item 3: If E is open, then every point of E is interior point. Since interior point is not boundary point, $E \cap \partial E = \emptyset$. For the converse, if $E \cap \partial E = \emptyset$, then for any $M \in E$, M is not boundary point, thus there exists $r > 0$ such that $B(M, r) \subset E$. This means that E is open.

Item 4: Notice that $\partial E = \partial E^c$ by definition. E is closed iff E^c is open iff $E^c \cap \partial E^c = \emptyset$, i.e. $E^c \cap \partial E = \emptyset$ by Item 3.

Item 5: By item 4, E is closed iff $\partial E \subset E$ which is obviously equivalent to the statement that E contains all of its accumulation points. $\square$

1.1.1 Connectivity of point set

With the notion of open and closed subset, we can define connectivity and path connectivity.

Definition 1.9 (Connected). A subset $E \subset \mathbf{R}^2$ is called *connected* if for any nontrivial decomposition $E = A \sqcup B$, either A contains some accumulation point of B , or B contains some accumulation point of A .

Characterization of “disconnectedness”: there exists a decomposition $E = A \sqcup B$ with $A, B \neq \emptyset$ such that $A \cap \partial B = \emptyset$ and $B \cap \partial A = \emptyset$.

Proposition 1.10. *Any connected subset of $\mathbf{R}$ is a interval.*

Proof. Suppose $E \subset \mathbf{R}$ is connected but there exists a triple of numbers $x_1 < x < x_2$ with $x_1, x_2 \in E$ but $x \notin E$. Then we have a nontrivial decomposition $E = (E \cap (-\infty, x)) \sqcup (E \cap (x, +\infty))$. It is easy to see any accumulation point of $E \cap (-\infty, x)$ in $\mathbf{R}$ is less than or equal to x , therefore cannot be in $E \cap (x, +\infty)$. Similarly, $E \cap (x, +\infty)$ does not contain any accumulation point of $E \cap (-\infty, x)$. This shows that E is disconnected by our above characterization. This is a contradiction. $\square$

Definition 1.11 (Path connected). If for any $P, Q \in E$, there exists a continuous curve in E connected P and Q , i.e. a map $\gamma = (\alpha, \beta) : [0, 1] \rightarrow E$ with α, β continuous function and $\gamma(0) = P$ and $\gamma(1) = Q$, then E is called path connected.

Theorem 1.12. *If E is path connected, then E is connected.*

Proof. Let $E = A \sqcup B$ with $A, B \neq \emptyset$. Take $P \in A$ and $Q \in B$. By assumption, we can take continuous $\gamma : [0, 1] \rightarrow E$ such that $\gamma(0) = P, \gamma(1) = Q$. Let

$$t_0 = \sup\{t \in [0, 1] \mid \gamma(t) \in A\}.$$

If $\gamma(t_0) \in B$, then by definition of supremum there exists t_n strictly increasing to t_0 and $\gamma(t_n) \in A$. If $\{\gamma(t_n) \mid n \in \mathbf{N}\}$ is a finite set, then we can take a subsequence t_{n_k} such that $\{\gamma(t_{n_k}) \mid k \in \mathbf{N}\}$ consists of a single point. By the continuity of γ , $\gamma(t_0) = \lim_{k \rightarrow \infty} \gamma(t_{n_k}) \in A$ which is contradiction. Therefore, $\{\gamma(t_n) \mid n \in \mathbf{N}\}$ is a infinite set so we can then take a subsequence t_{n_k} such that the point in the sequence $\{\gamma(t_{n_k})\}_{k \in \mathbf{N}}$ are all distinct. We thus conclude that $\gamma(t_0)$ is an accumulation point of A .

If $\gamma(t_0) \in A$, then $t_0 < 1$ and $\gamma(t) \notin A$ for any $t > t_0$. We can thus take a strictly decreasing sequence $t_n \rightarrow t_0$. By similar argument to the above case, we can prove that $\gamma(t_0)$ is an accumulation point of B . This implies that A contains an accumulation point of B . $\square$

An open and connected subset of $\mathbf{R}^2$ is called an open region. We can prove that any connected open set must be path connected (not true for general subset). By the connectivity (path connectivity) of $\mathbf{R}^2$, any subset of $\mathbf{R}^2$ which is both open and closed must be $\emptyset$ or $\mathbf{R}^2$.

Definition 1.13. If the continuous curve $\gamma : [\alpha, \beta] \rightarrow \mathbf{R}^2$ satisfies that $\gamma(t_1) \neq \gamma(t_2)$ if for any $\alpha \leq t_1 < t_2 \leq \beta$, then γ is called a *simple curve*. If $\gamma(\alpha) = \gamma(\beta)$, then the curve is called a *closed curve*.

Theorem 1.14 (Jordan curve theorem). *Any simple and closed curve γ on $\mathbf{R}^2$ divides $\mathbf{R}^2$ into two open regions. The bounded denoted by E^+ is called the interior region of γ , and the unbounded one denoted by E^- is called the exterior region of γ .*

The surface torus or higher genus surface shows that Jordan curve theorem is not apparently correct, it is a special property of the plane.

If the interior region of any simple closed curve in a region D is still inside D , then D is called simply connected. Otherwise, it is called multi-connected. Geometrically, simply connectedness means there is no holes in D .

2 Differentiation of multivariate functions

We've defined the meaning of a general map, injectivity, surjectivity, invertibility and compositions of maps in last semester, and now we specialize to the case of $\mathbf{R}^n$.

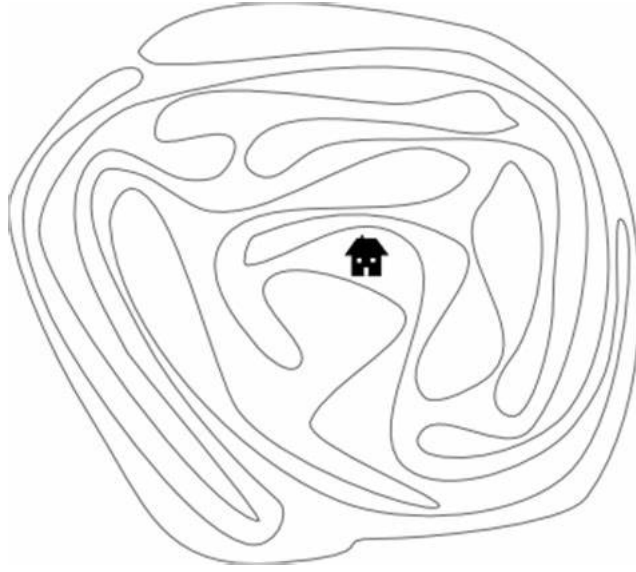


Figure 1: Jordan curve theorem

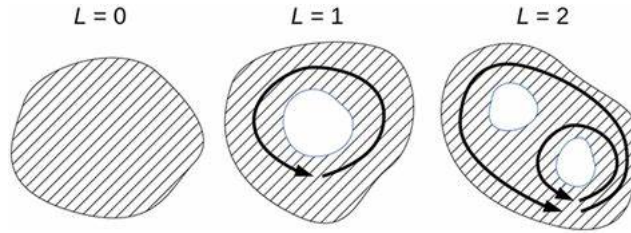


Figure 2: Simply connected and Multiply connected domain

Let $f : D(\subset \mathbf{R}^n) \longrightarrow \mathbf{R}$ be a map, then it is called a *multivariate function* of n variables. It can be written as $z = f(x_1, x_2, \dots, x_n)$ because point in $\mathbf{R}^n$ can be represented by its Cartesian coordinate.

How to represent such a function?

- Graph representation:

$$\mathcal{S} = \{(x_1, \dots, x_n, f(x_1, \dots, x_n)) \in \mathbf{R}^{n+1} | (x_1, \dots, x_n) \in D\}$$

is called the graph of the function $f : D \rightarrow \mathbf{R}$.

In the special case $n = 2$, this is a surface in $\mathbf{R}^3$, for instance $z = x^2 + y^2$ and $z = x^2 - y^2$.

- Level set representation: for any $c \in \mathbf{R}$,

$$f^{-1}(c) = \{(x_1, \dots, x_n) \in D | f(x_1, \dots, x_n) = c\} := L_c$$

is called the level set of f at value c .

Give examples by the quadratic functions under this two representations.

Isoheight line/Contour line, isothermic line....are widely used in geography.

2.1 Vector-valued function

To describe quantities in space which are not scalar, like electromagnetic field and force field or velocity field in a fluid, we need to consider more general vector-valued functions:

A map

$$\begin{aligned} \mathbf{f} : D(\subset \mathbf{R}^n) &\rightarrow \mathbf{R}^m \\ (x_1, \dots, x_n) &\mapsto (y_1, \dots, y_m) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)) \end{aligned}$$

is called a vector-valued function. We usually denote by $\mathbf{y} = \mathbf{f}(\mathbf{x})$. Each component $y_j = f_j(x_1, \dots, x_n)$ is a multivariate function of n variables.

Let us first consider the following examples:

- $n = 1, m = 3$: $\mathbf{r} : D(\subset \mathbf{R}) \rightarrow \mathbf{R}^3$ given by $\mathbf{r}(t) = (x(t), y(t), z(t))$ is a parameterized curve in $\mathbf{R}^3$;
- $n = 2, m = 3$, $\mathbf{r} : D(\subset \mathbf{R}^2) \rightarrow \mathbf{R}^3$ given by $\mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$ is a parameterized surface, for instance

$$\mathbf{r}(\theta, \varphi) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$$

for $(\theta, \varphi) \in [0, \pi] \times [0, 2\pi]$ parameterizes the unit sphere;

- $n = m$: a one-to-one vector-valued function $\mathbf{f} : D(\subset \mathbf{R}^n) \rightarrow \tilde{D}(\subset \mathbf{R}^n)$ is called a coordinate transformation. For instance, we can investigate the polar and spherical coordinate encountered before.

We can consider the composition of vector-valued functions: if $f : D(\subset \mathbf{R}^n) \rightarrow \mathbf{R}^m$ and $g : U(\subset \mathbf{R}^m) \rightarrow \mathbf{R}^k$ are two vector-valued functions such that $f(D) \subset U$, then the map $g \circ f$ is well defined, written as

$$\begin{aligned} g \circ f(x_1, \dots, x_n) &= (g_1(f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n)), \dots, \\ &\quad g_k(f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))), \end{aligned}$$

which can also be simply denoted by

$$\mathbf{z} = \mathbf{g}(\mathbf{f}(\mathbf{x})).$$

Example 2.1. Let $\gamma : [\alpha, \beta] \rightarrow D \subset \mathbf{R}^2$ be a curve and $\mathbf{f} : D \rightarrow \mathbf{R}^3$ be the graph parameterization of the function f , then $\mathbf{f} \circ \gamma$ is a parameterized curve on the graph of f .

Example 2.2. If $\Phi : \tilde{D} \rightarrow D \subset \mathbf{R}^2$ is a coordinate transformation, and $\mathbf{r} : D \rightarrow \mathbf{R}^3$ is a parameterized surface, then $\mathbf{r} \circ \Phi$ represents the same surface but with different parameterization.

2.2 Limit of multivariate functions

Definition 2.3 (Limit). Let $f : D(\subset \mathbf{R}^2) \rightarrow \mathbf{R}$, M_0 is an accumulation point of D . If for some $a \in \mathbf{R}$ we have that $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$, s.t. $\forall M \in D \cap B_-(M_0, \delta)$,

$$|f(M) - a| < \varepsilon,$$

then we say f converges to a as M converges to M_0 , and denote it by $\lim_{M \rightarrow M_0} f(M) = a$.

Thinking about the nested properties of disks and squares around a point, the following definition is equivalent

Definition 2.4. $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$, s.t. $\forall M = (x, y) \in D$ with $M \neq M_0$ satisfying $|x - x_0| < \delta, |y - y_0| < \delta$ ¹, we have

$$|f(M) - a| < \varepsilon.$$

This limit is also denoted by $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$, $\lim_{x \rightarrow x_0, y \rightarrow y_0} f(x, y)$ or $\lim_{\Delta x \rightarrow 0, \Delta y \rightarrow 0} f(x_0 + \Delta x, y_0 + \Delta y)$.

Similarly, we can define $\lim_{x \rightarrow +\infty, y \rightarrow +\infty} f(x, y) = a$ as

$\forall \varepsilon > 0, \exists R > 0$, s.t. $\forall (x, y)$ with $x > R, y > R$, we have $|f(x, y) - a| < \varepsilon$,

and $\lim_{x \rightarrow x_0, y \rightarrow +\infty} f(x, y) = a$ as

$\forall \varepsilon > 0, \exists R > 0, \delta > 0$, s.t. $\forall (x, y)$ with $|x - x_0| < \delta, y > R$, we have $|f(x, y) - a| < \varepsilon$.

We should remark that usually we define $\lim_{x \rightarrow \infty, y \rightarrow \infty} f(x, y) = a$ as

$\forall \varepsilon > 0, \exists R > 0$, s.t. $\forall (x, y)$ with $|x| > R$ or $|y| > R$ we have $|f(x, y) - a| < \varepsilon$.

In other words, we examine the values of the function outside any square $\{|x| < R, |y| < R\}$.

Example 2.5.

- Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = 0$;
- Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0$.

Proof. Using the inequality $x^2 y^2 \leq \frac{1}{2}(x^4 + y^2)|y|$ and $|x^3 + y^3| \leq (x^2 + y^2)(|x| + |y|)$. $\square$

The limit $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ should be distinguished with the so-called *iterative limit*

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y), \quad \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y),$$

¹Remark that there is some mistakes on the textbook at this point.

in general the existence of the three limits are different. The iterative limit only signifies a very special path to approach $(0, 0)$, for instance in the case $f(x, y) = \frac{xy}{x^2+y^2}$ the iterative limit $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y)$ basically depends only on the direction along the x -axis through $(0, 0)$. The real limit need to consider all possible (infinitely many) path of approaching $(0, 0)$.

Let us list the properties of limit of multivariate functions:

Theorem 2.6. *If $\lim_{M \rightarrow M_0} f(M)$, $\lim_{M \rightarrow M_0} g(M)$ both exist, then the limit $\lim_{M \rightarrow M_0} (f(M) \pm g(M))$, $\lim_{M \rightarrow M_0} f(M)g(M)$ exist and $\lim_{M \rightarrow M_0} \frac{f(M)}{g(M)}$ exists provided $\lim_{M \rightarrow M_0} g(M) \neq 0$.*

Theorem 2.7 (Heine Principle). *Suppose M_0 is an accumulation point of D and $f : D(\subset \mathbf{R}^2) \rightarrow \mathbf{R}$, then $\lim_{M \rightarrow M_0} f(M) = a$ if and only if for any point sequence $\{M_n\} \subset D$ with $M_n \neq M_0$ and $M_n \rightarrow M_0$ (as $n \rightarrow \infty$) there holds $\lim_{n \rightarrow \infty} f(M_n) = a$.*

Theorem 2.8. *Suppose f is defined in a punctured neighborhood $B_-(M_0, r)$ of M_0 , then $\lim_{M \rightarrow M_0} f(M) = a$ iff for any continuous curve $\gamma : [0, \delta) \rightarrow B_-(M_0, r)$ with $\gamma(0) = M_0$, $\gamma((0, \delta)) \subset B_-(M_0, r)$,*

$$\lim_{t \rightarrow 0^+} f(\gamma(t)) = a.$$

By replacing $|\cdot|$ with the distance function, we can define the limit of vector-valued multivariate functions.

Example 2.9. Study the limit of $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ as $(x, y) \rightarrow (0, 0)$.

Solution. Notice that along any line $y = kx$, we have

$$f(x, y) = \frac{kx^2}{x^4 + k^2 x^2} = \frac{kx}{k^2 + x^2},$$

and $\lim_{x \rightarrow 0} f(x, y) = 0$.

However, if we choose two different parabolic paths $y = k_1 x^2$ and $y = k_2 x^2$, then

$$f(x, y) = \frac{k}{1 + k^2},$$

and f has different limits along these two paths, therefore $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.

2.3 Continuity

Definition 2.10. Let $f : D(\subset \mathbf{R}^2) \rightarrow \mathbf{R}$ and $M_0 \in D$, we say f is continuous at M_0 if $\forall \varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that for any $M \in D \cap B(M_0, \delta)$, we have

$$|f(M) - f(M_0)| < \varepsilon.$$

It is equivalent to say $\lim_{M \rightarrow M_0} f(M) = f(M_0)$. If f is continuous at any $M \in D$, then f is called continuous on D .

Example 2.11. Study the continuity of the function $f(x, y) = \frac{xy}{x^2 + y^2}$. Graph it.

Solution. We know from last section that the limit of f at $(0, 0)$ does not exist, therefore it is not continuous at $(0, 0)$. Actually, the restriction of f to the circle $\gamma(\theta) = (\delta \cos \theta, \delta \sin \theta)$ is $f(\gamma(\theta)) = \cos \theta \sin \theta$ has oscillation 2 no matter how small δ is. This proves that f cannot be continuous at $(0, 0)$.

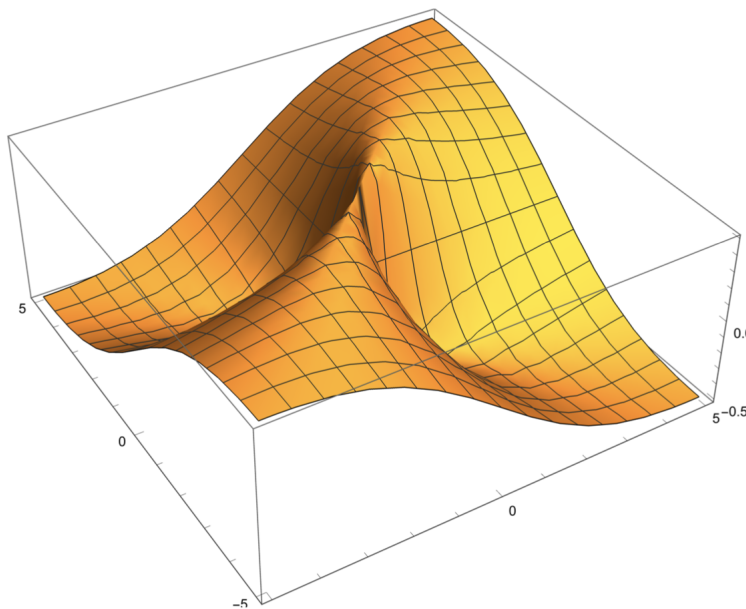


Figure 3: Geometrical meaning

Example 2.12. Define a function on the open region $D = \{(x, y) \in \mathbf{R}^2 | xy > -1\}$:

$$f(x, y) = \begin{cases} \log(1 + xy)^{\frac{1}{xy}}, & xy \neq 0; \\ 1, & xy = 0. \end{cases}$$

Proof. Since we have $\lim_{u \rightarrow 0} \log(1 + u)^{\frac{1}{u}} = 1$ (proved in last semester), for any $\varepsilon > 0$, there exists $\delta' = \delta'(\varepsilon) > 0$ such that for any $u \in (-\delta', \delta')$,

$$\left| \log(1 + u)^{\frac{1}{u}} - 1 \right| < \varepsilon.$$

For any $(0, y_0) \in D$ with $y_0 \neq 0$, take $\delta = \min\{|y_0|, \frac{\delta'}{2(|y_0|+1)}\}$, then for any $(x, y) \in D$ with $|x - 0| < \delta, |y - y_0| < \delta$,

$$|xy| \leq \delta \cdot 2|y_0| < \delta',$$

therefore we have

$$\left| \log(1 + xy)^{\frac{1}{xy}} - 1 \right| < \varepsilon.$$

This verifies the continuity at $(0, y_0)$. For $(x_0, 0) \in D$ with $x_0 \neq 0$, the argument is similar. For $(0, 0)$, we can simply take $\delta = \min\{1, \delta'\}$, then for any $|x| < \delta, |y| < \delta, |xy| < \delta^2 \leq \delta'$, and therefore

$$\left| \log(1 + u)^{\frac{1}{u}} - 1 \right| < \varepsilon.$$

Actually, later we can use the power series expansion $\log(1 + u)^{\frac{1}{u}} = 1 - \frac{u}{2} + \frac{u^2}{3} - \dots$ for $|u| < 1$ to show that $f(x, y)$ is smooth (even analytic) on the region D , and the critical point $(0, 0)$ is a saddle point. $\square$

Example 2.13. Define a function on $\mathbf{R}^2 \setminus \{(0, 0)\}$:

$$\theta(x, y) = \begin{cases} \arctan \frac{y}{x}, & x > 0, y \geq 0, \\ \frac{\pi}{2}, & x = 0, y > 0, \\ \arctan \frac{y}{x} + \pi, & x < 0, y > 0, \\ \arctan \frac{y}{x} + \pi, & x < 0, y < 0, \\ \frac{3\pi}{2}, & x = 0, y < 0, \\ \arctan \frac{y}{x} + 2\pi, & x > 0, y < 0. \end{cases}$$

where $\arctan : (-\infty, +\infty) \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ is the inverse of $\tan$ on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$. Study the continuity of the “angle function” $\theta(x, y)$.

Solution. It is easy to see the continuity of θ on the four open quadrants.

Near the critical $x = 0, y > 0$, we see that

$$\cos \theta(x, y) = \frac{x}{\sqrt{x^2 + y^2}},$$

and therefore

$$\theta(x, y) = \cos^{-1} \frac{x}{\sqrt{x^2 + y^2}},$$

which is the composition of two continuous function and thus continuous.

On the ray $x > 0, y = 0$, the function has a jump of 2π across the line from the first to the fourth quadrant:

In analogy to one variable functions, the notion of “uniform continuity” will be very useful in discussing the integrability of multivariate functions.

Definition 2.14 (uniform continuity). Let f be defined on $D \subset \mathbf{R}^2$, f is said to be uniformly continuous if $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$ s.t. $\forall M, M' \in D$ with $\rho(M, M') < \delta$, we have

$$|f(M) - f(M')| < \varepsilon.$$

Next, we list the essential properties of continuous functions:

- Theorem 2.15.** 1. The continuity is preserved under $+$, $-$, $\times$ and $/$ (provided the denominator is not 0);
2. The composition of continuous vector-valued functions is continuous;
3. (Intermediate Value Property) Let f be defined on a connected set $D \subset \mathbf{R}^2$ and f is continuous, then for any $M_1, M_2 \in D$, all values between $f(M_1), f(M_2)$ can be achieved. More generally, the image of any continuous function on a connected set is connected.
4. The maximum and minimum of continuous function on a bounded closed set can be achieved;
5. (Cantor's Theorem) Continuous function defined on bounded closed set is uniformly continuous.

Proof. The proofs of item 1 and 2 are left as exercises.

Proof of item 3: Argue by contradiction. Suppose $f(D)$ is not connected, then $f(D)$ can be decomposed as $E_1 \sqcup E_2$ such that $E_1, E_2 \neq \emptyset$ and no set contains the accumulation points of another. This induces a decomposition $D = f^{-1}(E_1) \sqcup f^{-1}(E_2)$ such that $f^{-1}(E_1), f^{-1}(E_2) \neq \emptyset$.

Let $M_n \in f^{-1}(E_1)$ be a sequence of distinct points such that $M_n \rightarrow M_\infty \in D$, then by continuity $f(M_n) \rightarrow f(M_\infty)$ (as $n \rightarrow \infty$). If $f(M_\infty) \notin E_1$, then $f(M_\infty)$ must be a limit point of E_2 as all $f(M_n) \in E_1$, this implies $f(M_\infty) \in E_2$ and therefore this is a contradiction since we assume E_2 does not contain any accumulation point of E_1 .

Now, if f takes value in $\mathbf{R}$, then the image $f(D)$ is a connected set of $\mathbf{R}$.

Proof of item 4 of item 5: exactly the same as one variable case in which Bolzano-Weierstrass Theorem is used. $\square$

Example 2.16. Discuss the limit

$$\lim_{(x,y) \rightarrow (0,0)} (1 + xy)^{\frac{1}{x+y}}.$$

Solution. Notice that

$$f(x, y) = e^{\frac{\log(1+xy)}{x+y}} = e^{\frac{\log(1+xy)}{xy} \cdot \frac{xy}{x+y}}.$$

As $(x, y) \rightarrow (0, 0)$, the first multiplicative factor converges to 1 by the elementary limit $\lim_{u \rightarrow 0} \frac{\log(1+u)}{u} = 1$. Therefore, the main problem comes from the term $\frac{xy}{x+y}$.

Along any line $y = kx$ (with $k \neq -1$), the function becomes $\frac{kx}{1+k}$ which is an $o(1)$ quantity, therefore converges to 0 as $(x, y) \rightarrow (0, 0)$. However, along the parabolic path $y = -x + kx^2$, we have

$$\frac{xy}{x+y} = \frac{x(-x + kx^2)}{kx^2} \rightarrow -\frac{1}{k}$$

as $x \rightarrow 0$. This means that along different parabolic paths approaching $(0, 0)$ the function takes different limits. Therefore the original limit does not exist.

Example 2.17. Discuss the continuity of the function

$$f(x, y) = \begin{cases} \frac{x-y}{x+y}, & x+y \neq 0 \\ 0, & x+y = 0 \end{cases}$$

Solution. For (x_0, y_0) not on the line $x + y = 0$, we use the properties of operations on continuous functions to get the continuity at (x_0, y_0) .

For (x_0, y_0) with $x_0 + y_0 = 0$ but $x_0 \neq 0$, then as $(x, y) \rightarrow (x_0, y_0)$ we have $x - y \rightarrow 2x_0$, and $x + y \rightarrow 0$ therefore $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x, y)$ does not exist.

For $(0, 0)$, we can take line $y = kx$ with $k \neq -1$, then $f(x, y) = \frac{1-k}{1+k}$, so the limit of $f(x, y)$ as $(x, y) \rightarrow (0, 0)$ is different along different lines, therefore the function is not continuous at $(0, 0)$.

3 Differentiation of multivariate functions

Recall the notion of first derivative for one variable function: it represents the instantaneous rate of change along the x -axis. We can generalize this meaning to multivariate functions. One important point is that the ratio

$$\frac{\Delta f}{\Delta M}$$

does not make sense since ΔM is a vector and Δf is a number/vector. The first way to overcome such difficulty is to look at the ratio along any coordinate axis.

Definition 3.1. Suppose $z = f(x, y)$ is defined in a neighborhood of $M_0 = (x_0, y_0)$. If the one variable function $f(x, y_0)$ has derivative at x_0 , we call this derivative as the first order partial derivative of f with respect to x at (x_0, y_0) , denoted by $\frac{\partial f}{\partial x}(x_0, y_0)$ or $f'_x(x_0, y_0)$, i.e.

$$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{d}{dx}|_{x=x_0} f(x, y_0).$$

Similarly, we can define $\frac{\partial f}{\partial y}(x_0, y_0)$.

If the first order partial derivatives of f exists at any point in D , then we get a first order partial derivative functions $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Geometrical meaning: $\frac{\partial f}{\partial x}(x_0, y_0)$ represents the slope of the tangent line to the graph of $f(x, y)$ in the x -direction.

Because of its definition, the rules of computation of partial derivatives is the same as the usual rules for one variable functions, for instance

$$\frac{\partial}{\partial x}(fg) = \frac{\partial f}{\partial x}g + f\frac{\partial g}{\partial x}.$$

Example 3.2. Find u'_x, u'_y, u'_z for the function $u(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$.

Example 3.3. Consider again the function

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

The function has $f'_x(0, 0) = f'_y(0, 0) = 0$ but it is not continuous at $(0, 0)$.

3.1 High order partial derivatives

Definition 3.4. Suppose $f(x, y)$ has first order partial derivative $\frac{\partial f}{\partial x}$ in a neighborhood of M_0 . If $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$ exists, we call it the second order partial derivative of f with respect to x . If $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$ exists, we call it the mixed second order partial derivative of f with respect to x and y .

Similarly, we can define $\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$ and $\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$.

Further, we can define higher order derivatives.

We have notations $\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$ and $\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$, $\frac{\partial^3 f}{\partial x \partial y^2}$ and so on....

In general, for $z = f(x_1, \dots, x_n)$, we can define

$$\frac{\partial^k f}{\partial x_{i_1} \cdots \partial x_{i_k}} = \frac{\partial}{\partial x_{i_1}} \left(\cdots \left(\frac{\partial f}{\partial x_{i_k}} \right) \cdots \right),$$

where $i_1, \dots, i_k \in \{1, 2, \dots, n\}$.

Since $\frac{\partial f}{\partial x}$ is the x -derivative of f along the x -line, the second order pure derivative $\frac{\partial^2 f}{\partial x^2} = \frac{d}{dx} \frac{\partial f}{\partial x}$ also depends only on the function $f(\cdot, y)$. It reduces to the usual 2^{nd} -order derivative.

We have

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2}(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x_0 + h, y_0) - \frac{\partial f}{\partial x}(x_0, y_0)}{h} \\ &= \lim_{h \rightarrow 0} \lim_{k \rightarrow 0} \frac{f(x_0 + h + k, y_0) - f(x_0 + h, y_0) - f(x_0 + k, y_0) + f(x_0, y_0)}{hk}. \end{aligned}$$

In particular, we can consider the second order difference

$$f(x_0 + h + k, y_0) - f(x_0 + h, y_0) - f(x_0 + k, y_0) + f(x_0, y_0) = \frac{\partial^2 f}{\partial x^2}(x_0 + \theta h + \eta k, y_0)hk.$$

Question 3.5. What's the relation between $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$?

$$\begin{aligned}
\frac{\partial^2 f}{\partial y \partial x}(x_0, y_0) &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) (x_0, y_0) \\
&= \lim_{\Delta y \rightarrow 0} \frac{f'_x(x_0, y_0 + \Delta y) - f'_x(x_0, y_0)}{\Delta y} \\
&= \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta y} \left\{ \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) + f(x_0, y_0)}{\Delta x} \right\} \\
&= \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{II(\Delta x, \Delta y)}{\Delta x \cdot \Delta y}.
\end{aligned}$$

For simplicity of notation, let $\Delta x = h, \Delta y = k$, then the mixed “second order difference” is

$$II(h, k) = f(x_0 + h, y_0 + k) - f(x_0, y_0 + k) - f(x_0 + h, y_0) + f(x_0, y_0).$$

If we denote $\varphi(x) = f(x, y_0 + k) - f(x, y_0)$, then by the Lagrange Mean Value Theorem,

$$\begin{aligned}
II(h, k) &= \varphi(x_0 + h) - \varphi(x_0) \\
&= \varphi'_x(x_0 + \theta_1 h)h \\
&= f'_x(x_0 + \theta_1 h, y_0 + k) - f'_x(x_0 + \theta_1 h, y_0) \\
&= \frac{\partial^2 f}{\partial y \partial x}(x_0 + \theta_1 h, y_0 + \eta_1 k)kh,
\end{aligned}$$

for some $\theta_1 = \theta_1(h, k), \eta_1 = \eta_1(h, k) \in (0, 1)$.

Similarly we can define $\psi(y) = f(x_0 + h, y) - f(x_0, y)$, then we can write $II(h, k) = \psi(y_0 + k) - \psi(y_0)$ and get

$$II(h, k) = \frac{\partial^2 f}{\partial x \partial y}(x_0 + \theta_2 h, y_0 + \eta_2 k)hk,$$

² for some $\theta_2 = \theta_2(h, k), \eta_2 = \eta_2(h, k) \in (0, 1)$.

For $h, k \neq 0$, we obtain then

$$\frac{\partial^2 f}{\partial y \partial x}(x_0 + \theta_1 h, y_0 + \eta_1 k) = \frac{\partial^2 f}{\partial x \partial y}(x_0 + \theta_2 h, y_0 + \eta_2 k).$$

If we assume the two mixed second order partial derivatives are both continuous at (x_0, y_0) then we can let $h, k \rightarrow 0$ to conclude that they are equal.

Theorem 3.6. Suppose $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ both exists in a neighborhood of M_0 and are continuous at M_0 , then $\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$.

²Geometric meaning of second order partial derivatives. You can try to develop the geometric meaning of other partial derivatives.

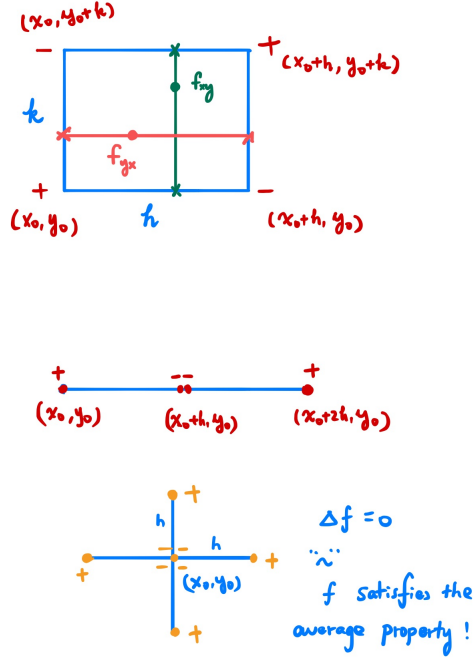


Figure 4: Difference quotient interpretation of partial derivatives

Example 3.7. Consider the example

$$f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2}(x^2 - y^2), & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0) \end{cases}$$

Solution. Let us first look at the first order partial derivatives:

$$\begin{aligned} \frac{\partial f}{\partial x}(0, 0) &= \lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = 0, \\ \frac{\partial f}{\partial x}(x, y) &= \frac{[2y(x^2 - y^2) + 2x \cdot 2xy](x^2 + y^2) - 2x \cdot 2xy(x^2 - y^2)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0), \\ \frac{\partial f}{\partial y}(0, 0) &= \lim_{y \rightarrow 0} \frac{f(0, y) - f(0, 0)}{y} = 0, \\ \frac{\partial f}{\partial y}(x, y) &= \frac{[2x(x^2 - y^2) - 2y \cdot 2xy](x^2 + y^2) - 2y \cdot 2xy(x^2 - y^2)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0). \end{aligned}$$

By the formulas, we can easily see that $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are both continuous everywhere. Moreover, f has arbitrary order of partial derivatives for $(x, y) \neq (0, 0)$.

Then, we can calculate the second order partial derivatives:

$$\begin{aligned}
\frac{\partial^2 f}{\partial x^2}(0, 0) &= \frac{d}{dx} \Big|_{x=0} \frac{\partial f}{\partial x}(x, 0) \\
&= \lim_{x \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x, 0) - \frac{\partial f}{\partial x}(0, 0)}{x} \\
&= \lim_{x \rightarrow 0} \left(\frac{[2y(x^2 - y^2) + 2x \cdot 2xy](x^2 + y^2) - 2x \cdot 2xy(x^2 - y^2)}{(x^2 + y^2)^2} \Big|_{(x, 0)} \right) / x \\
&= 0, \\
\frac{\partial^2 f}{\partial y^2}(0, 0) &= \frac{d}{dy} \Big|_{y=0} \frac{\partial f}{\partial y}(0, y) - \frac{\partial f}{\partial y}(0, 0) \\
&= \lim_{y \rightarrow 0} \left(\frac{[2x(x^2 - y^2) - 2y \cdot 2xy](x^2 + y^2) - 2y \cdot 2xy(x^2 - y^2)}{(x^2 + y^2)^2} \Big|_{(0, y)} \right) / y \\
&= 0, \\
\frac{\partial^2 f}{\partial x \partial y}(0, 0) &= \frac{d}{dx} \Big|_{x=0} \frac{\partial f}{\partial y}(x, 0) \\
&= \lim_{x \rightarrow 0} \frac{2x - 0}{x} \\
&= 2. \\
\frac{\partial^2 f}{\partial y \partial x}(0, 0) &= \frac{d}{dy} \Big|_{y=0} \frac{\partial f}{\partial x}(0, y) \\
&= \lim_{y \rightarrow 0} \frac{-2y - 0}{y} \\
&= -2.
\end{aligned}$$

Remark 3.8. A careful computation shows that

$$\Delta f = \frac{24xy(y^2 - x^2)}{(y^2 + x^2)^2}.$$

This means that Δf is bounded but not continuous. All the second order derivatives $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ are bounded. Moreover, the first order partial derivative $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are Lipschitz continuous. (This is consistent with the general Elliptic Regularity result from PDE theory!)

Example 3.9. Consider $f(x, y) = F(x) + G(y)$. Different regularity assumptions make different behavior of $f(x, y)$.

Example 3.10. If $\frac{\partial^2 f}{\partial x^2} \equiv 0$ and $\frac{\partial^2 f}{\partial y^2} \equiv 0$, then $f(x, y) = axy + bx + cy + d$ for some $a, b, c, d \in \mathbb{R}$.

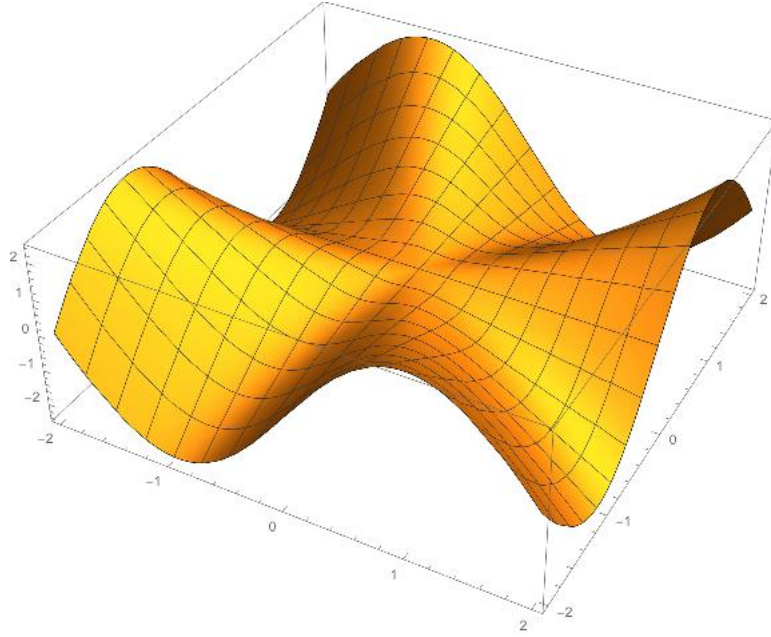


Figure 5: Graph of a function with non-commutative second order partial derivatives

Remark 3.11. It is pointed out by Yuze Song that we can assume a different inequivalent set of conditions to guarantee the exchange of mixed partial derivatives.

If we assume $\frac{\partial^2 f}{\partial x \partial y}$ is continuous at (x_0, y_0) , $\frac{\partial^2 f}{\partial y \partial x}$ exists at (x_0, y_0) , and moreover $\frac{\partial f}{\partial x}$ is horizontally continuous, then $\frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) = \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0)$.

Proof. Under the assumptions, we have

$$\begin{aligned} \frac{\partial^2 f}{\partial x \partial y}(x_0, y_0) &= \lim_{h, k \rightarrow 0} \frac{II(h, k)}{hk} = \lim_{k \rightarrow 0} \lim_{h \rightarrow 0} \frac{II(h, k)}{hk} = \lim_{k \rightarrow 0} \frac{\frac{\partial f}{\partial x}(0, k) - \frac{\partial f}{\partial x}(0, 0)}{k} \\ &= \frac{\partial^2 f}{\partial y \partial x}(x_0, y_0). \end{aligned}$$

□

4 Differentiability and Differential

The ratio “ $\frac{\Delta f}{\Delta M}$ ” can be understood as a map sending any variation in the domain ΔM to the corresponding variation in the image Δf . We can ask if the leading term of Δf with respect to ΔM is a linear function on ΔM , i.e. f can be approximated by a linear function near a point M .

Definition 4.1 (Differentiable). Let $f : D(\subset \mathbf{R}^n) \rightarrow \mathbf{R}$ be a multivariate function. If there exists a linear function $\mathcal{A}_{\mathbf{x}} : \mathbf{R}^n \rightarrow \mathbf{R}$ such that

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = \mathcal{A}_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|), \quad |\mathbf{h}| \rightarrow 0,$$

then we say f is differentiable at $\mathbf{x}$. And $\mathcal{A}_{\mathbf{x}}$ is called the differential of f at $\mathbf{x}$. We also denote $df|_{\mathbf{x}} = \mathcal{A}_{\mathbf{x}}$.

If f is differentiable at any point in D , then f is called differentiable on D .

Let us take n particular simple functions $f_i(\mathbf{x}) = x_i$, i.e. the coordinate functions. Then since

$$\Delta f_i(\mathbf{x} + \mathbf{h}) = x_i(\mathbf{x} + \mathbf{h}) - x_i(\mathbf{x}) = h_i,$$

which is definitely a linear function on $\mathbf{h}$. We know that f_i is differentiable, and $df_i|_{\mathbf{x}} : \mathbf{h} \mapsto h_i$. We can also write as $dx_i(\mathbf{h}) = h_i$.

For a general differentiable function f , since $df|_{\mathbf{x}}$ is a linear function on $\mathbf{h}$, there exists numbers $a_1, \dots, a_n$ (depending $\mathbf{x}$) such that for $\mathbf{h} = (h_1, \dots, h_n)$,

$$df|_{\mathbf{x}}(\mathbf{h}) = a_1 h_1 + \dots + a_n h_n.$$

Remembering that $dx_i(\mathbf{h}) = h_i$, we conclude that

$$df|_{\mathbf{x}} = a_1 dx_1 + \dots + a_n dx_n,$$

i.e. the differential of f is a linear combination of the basic differential $dx_1, \dots, dx_n$. What are the coefficients $a_1, \dots, a_n$ then?

By using the differentiability condition, we have

$$f(\mathbf{x} + t\mathbf{e}_i) - f(\mathbf{x}) = a_i t + o(|t|), \quad t \rightarrow 0.$$

This implies that

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{e}_i) - f(\mathbf{x})}{t} = a_i,$$

i.e. $\frac{\partial f}{\partial x_i}(\mathbf{x})$ exists and $a_i = \frac{\partial f}{\partial x_i}(\mathbf{x})$ for $i = 1, 2, \dots, n$.

Theorem 4.2. If f is differentiable at $\mathbf{x} = (x_1, \dots, x_n)$, then all of its first order partial derivatives $\frac{\partial f}{\partial x_i}(\mathbf{x})$ exists, and

$$df|_{\mathbf{x}} = \frac{\partial f}{\partial x_1}(\mathbf{x}) dx_1 + \dots + \frac{\partial f}{\partial x_n}(\mathbf{x}) dx_n.$$

Geometric meaning of differentiable: if $f(x, y)$ is differentiable at (x_0, y_0) , then

$$f(x, y) = f(x_0, y_0) + f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0) + o(\rho), \quad \rho \rightarrow 0,$$

this means the graph of f is tangential to the graph of the linear function $g(x, y) = f(x_0, y_0) + f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0)$.

Theorem 4.3. *If f is differentiable at $\mathbf{x}$, then f is continuous at $\mathbf{x}$.*

Proof. By the definition

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + a_1 h_1 + \cdots + a_n h_n + o(|\mathbf{h}|), \quad |\mathbf{h}| \rightarrow 0,$$

we have

$$|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x})| \leq |\mathbf{a}| \cdot |\mathbf{h}| + |o(|\mathbf{h}|)|,$$

then it is easily seen that $\lim_{\mathbf{h} \rightarrow 0} f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x})$. $\square$

The previous example $f(x, y) = \frac{2xy}{x^2 + y^2}$ shows that two partial derivatives exist do not imply the differentiability.

4.1 Differential and Partial derivatives

Because partial derivatives controls the rate of change of a function along the axial directions, information about partial derivatives in a neighborhood definitely say something about how a function varies in this neighborhood.

Theorem 4.4. *Suppose $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ exist in an open domain D .*

- *If $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are bounded on D , then f is (Lipschitz) continuous in D ;*
- *If $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are continuous in D , then f is differentiable in D .*

Proof. Item 1. Suppose $\left| \frac{\partial f}{\partial x} \right|, \left| \frac{\partial f}{\partial y} \right| \leq M$ on D , then we have

$$\begin{aligned} |f(x_0 + h, y_0 + k) - f(x_0, y_0)| &\leq |f(x_0 + h, y_0 + k) - f(x_0 + h, y_0)| \\ &\quad + |f(x_0 + h, y_0 + k) - f(x_0, y_0 + k)| \\ &\leq M(|h| + |k|). \end{aligned}$$

This implies that $\lim_{(h,k) \rightarrow (0,0)} f(x_0 + h, y_0 + k) = f(x_0, y_0)$.

Item 2. Fix a point (x_0, y_0) and consider the increment $\Delta x = h, \Delta y = k$, then

$$\Delta f = f'_x(x_0, y_0)h + f'_y(x_0, y_0)k + I,$$

where the error of Δf with its linear part is

$$I = (f'_x(x_0 + \theta h, y_0) - f'_x(x_0, y_0))h + (f'_y(x_0 + h, y_0 + \eta k) - f'_y(x_0, y_0))k.$$

We can then estimate the error I by the following:

$$\begin{aligned} \left| \frac{I}{\sqrt{h^2 + k^2}} \right| &\leq \frac{|h|}{\sqrt{h^2 + k^2}} |f'_x(x_0 + \theta h, y_0) - f'_x(x_0, y_0)| \\ &\quad + \frac{|k|}{\sqrt{h^2 + k^2}} |f'_h(x_0 + h, y_0 + \eta k) - f'_h(x_0, y_0)| \end{aligned}$$

By the continuity of f'_x and f'_h , we know that

$$\lim_{(h,k) \rightarrow (0,0)} \frac{I}{\sqrt{h^2 + k^2}} = 0,$$

which verifies the differentiability at (x_0, y_0) . $\square$

Remark 4.5. • From the proof we see that the statement is local in its nature, i.e. if we assume f'_x, f'_y is locally bounded near a point M_0 , then f is continuous at M_0 , and if f'_x, f'_y are both continuous at M_0 , then f is differentiability at M_0 .

- The condition f'_x and f'_y are continuous is just a sufficient condition for the differentiability, as can be seen from one variable function. Differentiability does not implies the continuity of its partial derivatives.

4.2 Directional derivatives and Gradient

Proposition 4.6. Suppose f and g are both differentiable at $\mathbf{x} \in D$, then

- For all $\lambda, \mu \in \mathbf{R}$, $\lambda f + \mu g$ is differentiable at $\mathbf{x}$, and

$$d(\lambda f + \mu g)|_{\mathbf{x}} = \lambda df|_{\mathbf{x}} + \mu dg|_{\mathbf{x}}.$$

- fg is differentiable at $\mathbf{x}$, and

$$d(fg)|_{\mathbf{x}} = df|_{\mathbf{x}} g(\mathbf{x}) + f(\mathbf{x}) dg|_{\mathbf{x}}.$$

- If $g(\mathbf{x}) \neq 0$, then $\frac{f}{g}$ is differentiable at $\mathbf{x}$, and

$$d\left(\frac{f}{g}\right)|_{\mathbf{x}} = \frac{df|_{\mathbf{x}} g(\mathbf{x}) - f(\mathbf{x}) dg|_{\mathbf{x}}}{g(\mathbf{x})^2}.$$

Proof. Let us only prove the Leibniz's rule.

$$\begin{aligned} fg(\mathbf{x} + \mathbf{h}) - fg(\mathbf{x}) &= (f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}))g(\mathbf{x} + \mathbf{h}) + f(\mathbf{x})(g(\mathbf{x} + \mathbf{h}) - g(\mathbf{x})) \\ &= (df|_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|))(g(\mathbf{x}) + \Delta g) + f(\mathbf{x})(dg|_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|)). \end{aligned}$$

The differentiability of g at $\mathbf{x}$ implies $\Delta g = o(1)$, we therefore can get the Leibniz's rule. $\square$

Now, assuming the differentiability of f at $\mathbf{x}$, then besides the axial directions (from which we obtain the partial derivatives) we can consider how does the function varies along an arbitrary direction $\mathbf{v}$.

$$f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x}) = df|_{\mathbf{x}}(t\mathbf{v}) + o(|t\mathbf{v}|) = t \cdot df|_{\mathbf{x}}(\mathbf{v}) + o(|t|), \quad t \rightarrow 0,$$

implies

$$\lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t} = df|_{\mathbf{x}}(\mathbf{v}).$$

This is called the derivative of f in the direction $\mathbf{v}$ at $\mathbf{x}$, or directional derivative for simplicity. We also denote by $\frac{\partial f}{\partial \mathbf{v}} = df|_{\mathbf{x}}(\mathbf{v})$. Analyzing further,

$$\begin{aligned} \frac{\partial f}{\partial \mathbf{v}}(\mathbf{x}) &= df|_{\mathbf{x}}(\mathbf{v}) = \frac{\partial f}{\partial x_1}(\mathbf{x})v_1 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{x})v_n \\ &= \text{grad } f(\mathbf{x}) \cdot \mathbf{v}, \end{aligned}$$

where $\text{grad } f(\mathbf{x}) = \frac{\partial f}{\partial x_1}(\mathbf{x})\mathbf{e}_1 + \cdots + \frac{\partial f}{\partial x_n}(\mathbf{x})\mathbf{e}_n := \nabla f(\mathbf{x})$ is called the *gradient* of f at $\mathbf{x}$.

Geometrically, if we take $\mathbf{v}$ to be unit, then $\frac{\partial f}{\partial \mathbf{v}}(\mathbf{x}) = |\nabla f(\mathbf{x})| \cos \theta$ where θ is the angle between $\nabla f(\mathbf{x})$ and $\mathbf{v}$. Therefore, $\nabla f(\mathbf{x})$ is the direction along which f increases most rapidly, and the level set of f is perpendicular to $\nabla f(\mathbf{x})$.

Remark 4.7. • We have $\frac{\partial f}{\partial \mathbf{e}_i} = \frac{\partial f}{\partial x_i}$ for $i = 1, 2, \dots, n$.

- Even we assume all directional derivatives exist at $\mathbf{x}$, we cannot conclude the differentiability of f at $\mathbf{x}$. For instance, consider the example

$$f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2} \sin \sqrt{x^2+y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

4.3 Differential of vector-valued functions

Now let us consider the differential of vector-valued functions. The condition is very easy to be generalized.

Let $\mathbf{f} : D(\subset \mathbf{R}^n) \rightarrow \mathbf{R}^m$ be a vector-valued function. If

$$\Delta \mathbf{y} = \mathcal{A}_{\mathbf{x}}(\Delta \mathbf{x}) + o(|\Delta \mathbf{x}|), \quad \Delta \mathbf{x} \rightarrow \mathbf{0},$$

for some linear map $\mathcal{A}_{\mathbf{x}} : \mathbf{R}^n \rightarrow \mathbf{R}^m$, then $\mathbf{f}$ is said to be differentiable at $\mathbf{x}$, and the linear map $\mathcal{A}_{\mathbf{x}}$ is called the differential of $\mathbf{f}$ at $\mathbf{x}$.

Under the standard bases $\mathbf{e}_1, \dots, \mathbf{e}_n$ of $\mathbf{R}^n$ and $\mathbf{e}_1, \dots, \mathbf{e}_m$ of $\mathbf{R}^m$ we can represent the linear map $\mathcal{A}_{\mathbf{x}} : \mathbf{R}^n \rightarrow \mathbf{R}^m$ by a $m \times n$ matrix $A = (a_{ij})$.

In terms of components, $\mathbf{y} = \mathbf{f}(\mathbf{x})$ is differentiable iff

$$\begin{aligned}\Delta y_1 &= (a_{11} \cdots a_{1n}) \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix} + o(|\Delta \mathbf{x}|) \\ &\vdots \\ \Delta y_m &= (a_{m1} \cdots a_{mn}) \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix} + o(|\Delta \mathbf{x}|)\end{aligned}$$

iff all component functions $f_1, \dots, f_m$ is differentiable. The question is how to represent $d\mathbf{f}|_x$?

$$\begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix} \mapsto \begin{pmatrix} df_1|_x(\Delta \mathbf{x}) \\ \vdots \\ df_m|_x(\Delta \mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \vdots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix} := J_{\mathbf{x}}(\mathbf{f}) \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \end{pmatrix}$$

where the coefficient matrix $J_{\mathbf{x}}(\mathbf{f})$ is called the *Jacobian matrix* of $\mathbf{f}$ at $\mathbf{x}$ (with respect to the standard bases).

We may call $d\mathbf{f}|_x(\mathbf{e}_i) = \frac{\partial f_1}{\partial x_i} \mathbf{e}_1 + \cdots + \frac{\partial f_m}{\partial x_i} \mathbf{e}_m$ the *partial derivative* of $\mathbf{f}$ in the direction x_i . Of course, now itself is a vector.

Remark 4.8. For $f : V \rightarrow \mathbb{R}$, we have that $df|_x \in V^*$. Elements in V^* are called *covectors*.

For $\mathbf{f} : U \rightarrow V$, we have that $d\mathbf{f}|_x \in U^* \otimes V$ for any $x \in U$. Elements of $U^* \otimes V$ are called *tensors*.

4.4 Differential of composition of functions

Suppose $\mathbf{f} : D(\subset \mathbb{R}^n) \rightarrow \mathbb{R}^m$ and $\mathbf{g} : U(\subset \mathbb{R}^m) \rightarrow \mathbb{R}^k$ are two vector valued functions and $\mathbf{f}(D) \subset U$. Suppose $\mathbf{f}$ is differentiable at $\mathbf{x}$ and $\mathbf{g}$ is differentiable at $\mathbf{y} = \mathbf{f}(\mathbf{x})$, i.e.

$$\begin{aligned}\mathbf{f}(\mathbf{x} + \mathbf{h}) &= \mathbf{f}(\mathbf{x}) + \mathcal{A}_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|), \quad \mathbf{h} \rightarrow \mathbf{0}, \\ \mathbf{g}(\mathbf{y} + \mathbf{l}) &= \mathbf{g}(\mathbf{y}) + \mathcal{B}_{\mathbf{y}}(\mathbf{l}) + o(|\mathbf{l}|), \quad \mathbf{l} \rightarrow \mathbf{0}.\end{aligned}$$

Plugging the first equation into the second, we obtain that

$$\begin{aligned}\mathbf{g}(\mathbf{f}(\mathbf{x} + \mathbf{h})) &= \mathbf{g}(\mathbf{y}) + \mathcal{B}_{\mathbf{y}}(\mathcal{A}_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|)) + o(1) \cdot (\mathcal{A}_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|)) \\ &= \mathbf{g}(\mathbf{y}) + \mathcal{B}_{\mathbf{y}} \circ \mathcal{A}_{\mathbf{x}}(\mathbf{h}) + o(|\mathbf{h}|), \quad \mathbf{h} \rightarrow \mathbf{0}.\end{aligned}$$

This means that $\mathbf{g} \circ \mathbf{f}$ is differentiable at $\mathbf{x}$, and

$$d(\mathbf{g} \circ \mathbf{f})|_{\mathbf{x}} = d\mathbf{g}|_{\mathbf{y}=\mathbf{f}(\mathbf{x})} \circ d\mathbf{f}|_{\mathbf{x}}. \quad (4.1)$$

This is the chain rule for the differentiation of composition of vector-valued functions.

Fixing the standard bases $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, $\{\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_m\}$ and $\{\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_k\}$ for $\mathbf{R}^n$, $\mathbf{R}^m$ and $\mathbf{R}^k$ respectively. Then $d\mathbf{f}|_{\mathbf{x}}$ is represented by $J_{\mathbf{x}}(\mathbf{f}) = \left(\frac{\partial y_i}{\partial x_j} \right)_{1 \leq i \leq m, 1 \leq j \leq n}$, similarly $d\mathbf{g}|_{\mathbf{y}}$ is represented by $J_{\mathbf{y}}(\mathbf{g}) = \left(\frac{\partial z_p}{\partial y_q} \right)_{1 \leq p \leq k, 1 \leq q \leq m}$. Then, we obtain that

$$J_{\mathbf{x}}(\mathbf{g} \circ \mathbf{f}) = J_{\mathbf{y}=\mathbf{f}(\mathbf{x})}(\mathbf{g}) \cdot J_{\mathbf{x}}\mathbf{f} = \left(\frac{\partial z_p}{\partial y_q} \right) \left(\frac{\partial y_i}{\partial x_j} \right) = \left(\sum_{q=1}^m \frac{\partial z_p}{\partial y_q} \frac{\partial y_q}{\partial x_j} \right)_{1 \leq p \leq k, 1 \leq j \leq n}.$$

Theorem 4.9 (Chain Rule for partial derivatives).

$$\frac{\partial z_p}{\partial x_j}(\mathbf{x}) = \sum_{q=1}^m \frac{\partial z_p}{\partial y_q}(\mathbf{f}(\mathbf{x})) \frac{\partial y_q}{\partial x_j}(\mathbf{x}).$$

In the special case when $n = m$, the determinant of the Jacobian matrix is called the *Jacobian determinant*,

$$\det J_{\mathbf{x}}(\mathbf{f}) = \frac{\partial(y_1, \dots, y_n)}{\partial(x_1, \dots, x_n)},$$

this notion will be crucially used in integration theory later.

Example 4.10. Suppose $u = f(x, y, z)$, $y = \varphi(x, r)$, $z = \psi(x, y, r)$. Find $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial r}$.

Solution. Recognizing the composition structure: u is a two variable function sending (x, r) to $(x, \varphi(x, r), \psi(x, \varphi(x, r), r))$, therefore by the Chain rule:

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial x} + \frac{\partial f}{\partial z} \left(\frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} \frac{\partial \varphi}{\partial x} \right). \\ \frac{\partial u}{\partial r} &= \frac{\partial f}{\partial y} \frac{\partial \varphi}{\partial r} + \frac{\partial f}{\partial z} \left(\frac{\partial \psi}{\partial y} \frac{\partial \varphi}{\partial r} + \frac{\partial \psi}{\partial r} \right). \end{aligned}$$

Example 4.11. Let $\xi = \xi(u, v)$, $\zeta = \zeta(u, v)$ and $u = u(x, y)$, $v = v(x, y)$, then

$$\begin{aligned} \frac{\partial \xi}{\partial x} &= \frac{\partial \xi}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \xi}{\partial v} \frac{\partial v}{\partial x} \\ \frac{\partial \xi}{\partial y} &= \frac{\partial \xi}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \xi}{\partial v} \frac{\partial v}{\partial y}. \end{aligned}$$

Example 4.12. Let $u = f(x, y)$ and $x = r \cos \theta$, $y = r \sin \theta$ and u is differentiable. Show that

$$\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2.$$

Example 4.13. Let $z = f\left(\log x + \frac{1}{y}\right)$ and f be differentiable, prove that $x \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = 0$.

5 Implicit Function Theorem and Inverse Function Theorem

The previous discussion about quadric surface poses the question how to study the zero set of a general function. For instance, how does one variable depend on other variables for the zero set?

Two difficulties:

- The example is $\sin(x + y) + 2x + y = 0$, $x, y \in \mathbf{R}$. It is difficult to solve explicit $y = y(x)$, how do we understand the dependence of y in terms of x , for instance $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$?
- The example is $x^2 + y^2 - 1 = 0$, $x, y \in \mathbf{R}$. Globally, y is not a function on x (one x may corresponds to several y , and some x does not correspond to any y).

5.1 Two variables case

Definition 5.1 (Implicit function). Let F be defined in an open domain on $\mathbf{R}^2$, let $C = \{(x, y) \in D | F(x, y) = 0\}$. Take $(x_0, y_0) \in C$, if there exists a neighborhood $I \times J \subset D$ of (x_0, y_0) (where I, J are interval) such that: $\forall x \in I, \exists! y \in J$ s.t. $F(x, y) = 0$, then the correspondence $I \rightarrow J$ sending x to the y is called an *implicit function* determined by $F(x, y)$ in the neighborhood $I \times J$ of (x_0, y_0) .

Example 5.2. Let $F(x, y) = x^2 + y^2 - 1$, then $C = \{(x, y) \in \mathbf{R}^2 | x^2 + y^2 - 1 = 0\}$. Study the possible implicit functions defined by F .

Solution. 1. For any $y_0 > 0$, we can take $I \times J = (-1, 1) \times (0, 2)$, then $F = 0$ defines an implicit function in this neighborhood of (x_0, y_0) . Actually, this implicit function has a explicit formula: $y = \sqrt{1 - x^2}$.

2. Similarly for $y_0 < 0$, $F = 0$ defines an implicit function in the neighborhood $(-1, 1) \times (-2, 0)$, it also has explicit formula $y = -\sqrt{1 - x^2}$.

3. For $(x_0, y_0) = (-1, 0)$. Then in any rectangular neighborhood of (x_0, y_0) , we cannot define an implicit function $y = y(x)$. However, if we take $I \times J = (-2, 0) \times (-1, 1)$, then we can define an implicit function $x = x(y)$ from $J \rightarrow I$. It has explicit formula $x = -\sqrt{1 - y^2}$.

Example 5.3. Consider the function $F(x, y) = x^2 - y^2$ near the point $(x_0, y_0) = (0, 0)$.

On any neighborhood $I \times J$ of $(0, 0)$, the set $\{F = 0\}$ cannot be represented as the graph of an implicit function $y = y(x)$ or $x = x(y)$. Thus, there is no implicit function near $(0, 0)$.

Question: Under what condition does F determines an implicit function near (x_0, y_0) ?

Theorem 5.4 (Implicit Function Theorem).³ Let $D \subset \mathbf{R}^2$ be an open domain, $M_0 = (x_0, y_0) \in D$. Let F be defined on D and satisfy

- $F \in C^1(D)$, i.e. continuously differentiable;
- $F(x_0, y_0) = 0$;
- $F'_y(x_0, y_0) \neq 0$,

then there exists a rectangular neighborhood $I \times J$ of (x_0, y_0) such that $F(x, y) = 0$ determines an implicit function $y = y(x)$ in $I \times J$. In other words, $\forall x \in I, \exists y = y(x) \in J$ such that $F(x, y) = 0$. Moreover, $y = y(x)$ is continuously differentiable, and

$$\frac{dy}{dx} = -\frac{F'_x(x, y)}{F'_y(x, y)}.$$

Example 5.5. Consider $F(x, y) = x^2 + y^2 - 1$.

Solution. Firstly, $F'_y = 2y$. Suppose $F(x_0, y_0) = 0$ such that $y_0 > 0$, then the Implicit Function Theorem implies there exists a rectangular neighborhood $I \times J$ of (x_0, y_0) such that $F = 0$ defines an implicit function $y = y(x)$ in $I \times J$.

By the theorem, we also have

$$\frac{dy}{dx} = -\frac{F'_x(x, y)}{F'_y(x, y)} = -\frac{2x}{2y} = -\frac{x}{y}.$$

We do not need to solve the derivative in terms of x all explicitly. It can certainly contains the implicit function y itself.

Example 5.6. Consider the example $F(x, y) = \sin(x + y) + 2x + y$.

Solution. Since $F'_x = \cos(x + y) + 2, F'_y = \cos(x + y) + 1$.

For any $(x, y) \in \mathbf{R}^2$, $F'_x > 0$, this implies that for any (x_0, y_0) , we can apply the Implicit Function Theorem to obtain a neighborhood $I \times J$ such that $F = 0$ determines an implicit function $x = x(y)$. We claim that I can be taken as $\mathbf{R}$.

Proof of the claim: for any $y \in \mathbf{R}$, $\lim_{x \rightarrow -\infty} F(x, y) = -\infty, \lim_{x \rightarrow +\infty} F(x, y) = +\infty$ and $F(\cdot, y)$ is continuous about $x \in \mathbf{R}$, therefore by the Intermediate Value Theorem, there exists $x \in \mathbf{R}$ such that $F(x, y) = 0$. On the other hand, $F'_x > 0$ implies $F(\cdot, y)$ is strictly increasing in x , thus there exists a unique $x \in \mathbf{R}$ with $F(x, y) = 0$.

Now, let us consider the possibility of determining implicit function $y = y(x)$. The equations

$$\begin{cases} F(x, y) &= 0 \\ F'_y(x, y) &= 0 \end{cases}$$

³The difficulty in understanding this theorem is about the meaning of “existence of implicit function”. It is originated from the work of Descartes, Bernoulli, Euler and appeared in the modern form in the lecture notes of U. Dini around 1880.

has solution $M_k = (-(2k+1)\pi, (4k+2)\pi)$ ($k \in \mathbf{Z}$). Therefore, near any $(x, y) \in C \setminus \{M_k | k \in \mathbf{Z}\}$ the Implicit Function Theorem determines a implicit function $y = y(x)$. Moreover,

$$\frac{dy}{dx} = -\frac{\cos(x+y)+2}{\cos(x+y)+1}.$$

Denote the interval $I_k = (-(2k+1)\pi, -(2k-1)\pi)$. We can prove that for any $(x, y) \in C$ with $x \in I_k$, the neighborhood claimed in the Implicit Function Theorem can be taken to be I_k (*Prove by yourself!*). Let us denote this implicit function as φ_k . Prove that all these φ_k 's can be pieced together to give a global continuous implicit function $y = \varphi(x)$ on $\mathbf{R}$. Of course, φ is not differentiable at M_k 's.

Example 5.7 (Folium of Descartes). Use Implicit Function Theorem to study the curve:

$$C : F(x, y) = x^3 + y^3 - 3axy.$$

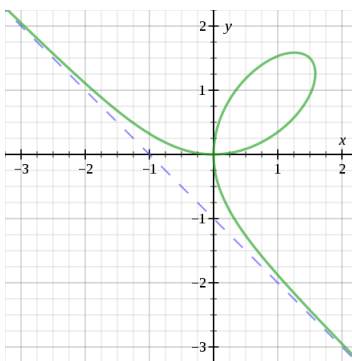


Figure 6: Folium of Descartes, 1683

Solution. The curve has a parametric equation

$$(x(t), y(t)) = \left(\frac{3at}{1+t^3}, \frac{3at^2}{1+t^3} \right), \quad t \in \mathbf{R} \setminus \{-1\}.$$

Solving the equations

$$\begin{cases} F(x, y) &= x^3 + y^3 - 3axy = 0 \\ F'_y(x, y) &= 3y^2 - 3ax = 0 \end{cases},$$

we obtain $(x, y) = (0, 0)$ or $(\sqrt[3]{4a}, \sqrt[3]{2a})$. Similarly, solving the equations

$$\begin{cases} F(x, y) &= x^3 + y^3 - 3axy = 0 \\ F'_x(x, y) &= 3x^2 - 3ay = 0 \end{cases},$$

we obtain $(x, y) = (0, 0)$ or $(x, y) = (\sqrt[3]{2a}, \sqrt[3]{4a})$.

1. For $(x_0, y_0) \in C \setminus \{(0, 0), (\sqrt[3]{4}a, \sqrt[3]{2}a)\}$, we can solve an implicit function $y = y(x)$ in a neighborhood of (x_0, y_0) , and

$$\frac{dy}{dx} = -\frac{x^2 - ay}{y^2 - ax}.$$

2. For $(x_0, y_0) \in C \setminus \{(0, 0), (\sqrt[3]{2}a, \sqrt[3]{4}a)\}$, we can solve an implicit function $x = x(y)$ in a neighborhood of (x_0, y_0) , and

$$\frac{dx}{dy} = -\frac{y^2 - ax}{x^2 - ay}.$$

3. For $(x_0, y_0) = (0, 0)$, there does not exist implicit function in any neighborhood.

Solving the following equations

$$\begin{cases} \frac{dy}{dx} = 0 \\ F(x, y) = 0. \end{cases}$$

we obtain that $\sqrt[3]{2}a$ is a critical point of the implicit function near the point $(x, y) = (\sqrt[3]{2}a, \sqrt[3]{4}a)$. We can then look at the second derivative of the implicit function:

$$\begin{aligned} \frac{d^2y}{dx^2} \Big|_{(\sqrt[3]{2}a, \sqrt[3]{4}a)} &= \frac{d}{dx} \Big|_{x=\sqrt[3]{2}a} \left(-\frac{x^2 - ay}{y^2 - ax} \right) \\ &= -\frac{\left(2x - a \frac{dy}{dx} \right) (y^2 - ax) - (x^2 - ay) \left(2y \frac{dy}{dx} - a \right)}{(y^2 - ax)^2} \Big|_{(\sqrt[3]{2}a, \sqrt[3]{4}a)} \\ &= -\frac{2\sqrt[3]{2}a(\sqrt[3]{16}a^2 - \sqrt[3]{2}a^2)}{(\sqrt[3]{16}a^2 - \sqrt[3]{2}a^2)^2} \\ &= -\frac{2\sqrt[3]{2}}{\sqrt[3]{16} - \sqrt[3]{2}} \frac{1}{a} \\ &< 0. \end{aligned}$$

This implies that $y = \sqrt[3]{4}a$ is a local maximum of the implicit function.

Proof of the Implicit Function Theorem.

Step 1: **The existence of the implicit function.** Since we assume $F(x_0, y_0) = 0$ and $F'_y(x_0, y_0) \neq 0$. WLOG, we assume $F'_y(x_0, y_0) > 0$. Since we assume F is continuously differentiable, there exists a rectangular neighborhood $I' \times J = [a', b'] \times [c, d]$ of (x_0, y_0) such that $F'_y|_{I' \times J} > 0$. Since $F(x_0, \cdot)$ is strictly increasing on $[c, d]$ and F is continuous on $I' \times J$, there exists $I = [a, b] \subset I'$ such that $F|_{I \times \{c\}} < 0, F|_{I \times \{d\}} > 0$. Then, by the continuity of the function $F(x, \cdot)$ on $[c, d]$ for any fixed $x \in I$, and the Intermediate Value Theorem,

we know that there exists $y \in [c, d]$ such that $F(x, y) = 0$. Because $F(x, \cdot)$ is strictly increasing in the y -variable, such y corresponding to x is unique. So this defines an implicit function $y : I \times J$ sending x to the unique $y \in J$ such that $F(x, y) = 0$.

Step 2: **Continuity of the implicit function.** Let $x' \in I$, take any sequence $\{x_n\}_{n \in \mathbf{N}}$ such that $x_n \rightarrow x'$, then since $\{y_n = y(x_n)\}_{n \in \mathbf{N}} \subset J$ we can use Bolzano-Weierstrass Theorem to conclude that there exists a sequence $\{x_{n_k}\}_{k \in \mathbf{N}}$ such that $y_{n_k} \rightarrow y' \in J$ as $k \rightarrow +\infty$. By the continuity of F , we know that $F(x', y') = \lim_{k \rightarrow +\infty} F(x_{n_k}, y_{n_k}) = 0$. By the uniqueness of y (satisfying $F(x', y) = 0$) from last step, we know that $y' = y(x')$. We've shown that for any $\{x_n\}$ converging to x' , there exists a subsequence $\{x_{n_k}\}$ such that $y(x_{n_k}) \rightarrow y'$. This implies (by the Heine Principle) that the whole sequence $\{y_n = y(x_n)\}$ converges to $y' = y(x')$. Precisely the continuity.

Step 3: **Differentiability of the implicit function.** We only need to discuss the differentiability of $y = y(x)$ at x_0 . Since

$$\begin{aligned} 0 &= F(x_0 + \Delta x, y_0 + \Delta y) - F(x_0, y_0) \\ &= F'_x(x_0 + \theta \Delta x, y_0) \Delta x + F'_y(x_0 + \Delta x, y_0 + \eta \Delta y) \Delta y \end{aligned}$$

for some $\theta, \eta \in (0, 1)$. We then have we have

$$\frac{\Delta y}{\Delta x} = - \frac{F'_x(x_0 + \theta \Delta x, y_0)}{F'_y(x_0 + \Delta x, y_0 + \eta \Delta y)}.$$

And therefore

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = - \frac{F'_x(x_0, y_0)}{F'_y(x_0, y_0)}$$

by the continuity of F'_x and F'_y . This verifies that $\frac{dy}{dx} = - \frac{F'_x(x, y)}{F'_y(x, y)}$ (with $y = y(x)$ is the implicit function). Moreover, $\frac{dy}{dx}$ is continuous on I , so $y = y(x) \in C^1(I)$. □

Example 5.8. The example $F(x, y) = y^3 - x$ near $(0, 0)$ admits implicit functions $x = y^3$ and $y = x^{\frac{1}{3}}$. However, $F'_x(0, 0) = F'_y(0, 0) = 0$.

Remark 5.9. If the $F'_x(0, 0) = 0$, we may not be able to perturb the problem to get a solution $x = x(y)$ for y near 0.

For a concrete example, consider the critical point problem of the functions:

1. $f(x, y) = y^2 \left(x - \sin \frac{1}{y} \right)^2$ where $y \in (-\epsilon, \epsilon)$ is the variation parameter.

2.

$$g(x, y) = y \left(x - \frac{1}{y} \right)^2.$$

3.

$$h(x, y) = y(x - y)^2.$$

- For $y = 0$, the function $f(x, 0) : \mathbb{R} \rightarrow \mathbb{R}$ has a whole $\mathbb{R}$ -worth of critical points;
- For each $y \neq 0$, the function $f(x, y) : \mathbb{R} \rightarrow \mathbb{R}$ only has a unique critical point $x = \sin \frac{1}{y}$.
- The above two facts implies that one cannot simply perturb a critical point x_0 of $f(\cdot, 0)$ to a nearby critical point of $f(\cdot, y)$ for nearby y .

5.2 General form of Implicit Function Theorem

Theorem 5.10 (Implicit Function Theorem). *Let $D \subset \mathbf{R}^n \times \mathbf{R}^m$ be an open domain and $M_0 = (\mathbf{x}^0, \mathbf{y}^0)$. If the vector-valued function $\mathbf{F} : D \rightarrow \mathbf{R}^m$ satisfies*

- $\mathbf{F} \in C^1(D)$;
- $\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0) = \mathbf{0}$;
- *The partial differential $d_{\mathbf{y}}\mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)} : \mathbf{R}^m \rightarrow \mathbf{R}^m$ is invertible, i.e. the corresponding partial Jacobian matrix*

$$J_{\mathbf{y}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0) = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \vdots & \cdots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \cdots & \frac{\partial F_m}{\partial y_m} \end{pmatrix} (\mathbf{x}^0, \mathbf{y}^0)$$

is invertible.

Then there exists a neighborhood I of $\mathbf{x}^0$ in $\mathbf{R}^n$ and a neighborhood J of $\mathbf{y}^0$ in $\mathbf{R}^m$ such that the equation $\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$ determines an implicit function $\mathbf{y} : I \rightarrow J$, and moreover $\mathbf{y} = \mathbf{y}(\mathbf{x})$ is continuously differentiable and

$$d\mathbf{y} = -(d_{\mathbf{y}}\mathbf{F})^{-1} \circ (d_{\mathbf{x}}\mathbf{F})$$

i.e. $J_{\mathbf{x}}\mathbf{y} = -\left(J_{\mathbf{y}}(\mathbf{F})\right)^{-1} \cdot J_{\mathbf{x}}(\mathbf{F})$ or more explicitly

$$\begin{pmatrix} dy_1 \\ \vdots \\ dy_m \end{pmatrix} = - \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \vdots & \cdots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \cdots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}^{-1} \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \cdots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_m} \end{pmatrix} \begin{pmatrix} dx_1 \\ \vdots \\ dx_n \end{pmatrix}.$$

The Step 1 about the existence of the implicit function will be given in Mathematical Analysis III. We need to consider a *contracting mapping*

$$G_{\mathbf{x}} : \mathbf{y} \mapsto \mathbf{y} - (d_{\mathbf{y}}\mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)})^{-1} (\mathbf{F}(\mathbf{x}, \mathbf{y})),$$

for all $\mathbf{x}$ sufficiently close to $\mathbf{x}^0$. The *fixed point* of the map $\mathbf{G}_{\mathbf{x}}$ gives the unique $\mathbf{y}$ such that $\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$. At the moment, we can sketch the proof of Step 2 and Step 3.

Step 2: This only depends on Bolzano-Weierstrass Theorem and Heine Principle. Therefore the proof in the $n = m = 1$ case carries over. We establish the continuity of the implicit function.

Step 3: According to $\mathbf{F}(\mathbf{x}^0 + \Delta\mathbf{x}, \mathbf{y}^0 + \Delta\mathbf{y}) - \mathbf{F}(\mathbf{x}^0, \mathbf{y}^0) = \mathbf{0}$, we have

$$\begin{aligned} 0 &= F_i(\mathbf{x}^0 + \Delta\mathbf{x}, \mathbf{y}^0 + \Delta\mathbf{y}) - F_i(\mathbf{x}^0, \mathbf{y}^0) \\ &= F_i(x_1^0 + \Delta x_1, \dots, x_n^0 + \Delta x_n, y_1^0 + \Delta y_1, \dots, y_m^0 + \Delta y_m) - F_i(x_1^0, \dots, x_n^0, y_1^0, \dots, y_m^0) \\ &= \sum_{k=1}^m \frac{\partial F_i}{\partial y_k}(x_1^0 + \Delta x_1, \dots, x_n^0 + \Delta x_n, y_1^0 + \Delta y_1, \dots, y_k^0 + \eta_k \Delta y_k, y_{k+1}^0, \dots, y_m^0) \Delta y_k \\ &\quad + \sum_{l=1}^n \frac{\partial F_i}{\partial x_l}(x_1^0 + \Delta x_1, \dots, x_l^0 + \theta_l \Delta x_l, x_{l+1}^0, \dots, x_n^0, y^0, \dots, y_m^0) \Delta x_l, \quad i = 1, 2, \dots, m. \end{aligned}$$

Writing this in terms of matrix multiplication,

$$\begin{pmatrix} \frac{\partial F_1}{\partial x_1}(M_{11}) & \cdots & \frac{\partial F_1}{\partial x_n}(M_{1n}) & \frac{\partial F_1}{\partial y_1}(N_{11}) & \cdots & \frac{\partial F_1}{\partial y_m}(N_{1m}) \\ \vdots & \cdots & \vdots & \vdots & \cdots & \vdots \\ \frac{\partial F_m}{\partial x_1}(M_{m1}) & \cdots & \frac{\partial F_m}{\partial x_n}(M_{mn}) & \frac{\partial F_m}{\partial y_1}(N_{m1}) & \cdots & \frac{\partial F_m}{\partial y_m}(N_{mm}) \end{pmatrix} \begin{pmatrix} \Delta x_1 \\ \vdots \\ \Delta x_n \\ \Delta y_1 \\ \vdots \\ \Delta y_m \end{pmatrix} = \mathbf{0}_m,$$

for appropriate points M_{ij}, N_{kl} in the rectangular box generated by $(\mathbf{x}^0, \mathbf{y}^0)$ and $(\mathbf{x}^0 + \Delta\mathbf{x}, \mathbf{y}^0 + \Delta\mathbf{y})$. This can be expanded in terms of the multiplication of blocked matrices:

$$(A \mid B) \begin{pmatrix} \Delta\mathbf{x} \\ \Delta\mathbf{y} \end{pmatrix} = A\Delta\mathbf{x} + B\Delta\mathbf{y} = \mathbf{0}_m.$$

According to the assumption of continuously differentiability, there exists a sufficient small neighborhood I of $\mathbf{x}^0$ and J of $\mathbf{y}^0$ such that for any $N_{ij} \in I \times J$, the matrix B is invertible (think about why). Therefore, we have

$$\Delta\mathbf{y} = -B^{-1}A\Delta\mathbf{x}.$$

Near the point $(\mathbf{x}^0, \mathbf{y}^0)$, $A = D_{\mathbf{x}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0) + o(1)$, $B = D_{\mathbf{y}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0) + o(1)$ and $B^{-1} = (D_{\mathbf{y}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0))^{-1} + o(1)$ (**think about why**). We therefore have

$$\Delta\mathbf{y} = -(D_{\mathbf{y}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0))^{-1} \cdot D_{\mathbf{x}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0)\Delta\mathbf{x} + o(\Delta\mathbf{x}), \quad \Delta\mathbf{x} \rightarrow \mathbf{0},$$

which means that $\mathbf{y} = \mathbf{y}(\mathbf{x})$ is differentiable at $\mathbf{x}^0$ with

$$J_{\mathbf{x}^0}\mathbf{y} = -(D_{\mathbf{y}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0))^{-1} \cdot D_{\mathbf{x}}\mathbf{F}(\mathbf{x}^0, \mathbf{y}^0).$$

By the uniqueness of the implicit function in this neighborhood $I \times J$, we actually know that

$$J_x \mathbf{y} = -(D_{\mathbf{y}} \mathbf{F}(\mathbf{x}, \mathbf{y}))^{-1} \cdot D_x \mathbf{F}(\mathbf{x}, \mathbf{y}), \quad \forall \mathbf{x} \in I.$$

By the continuity of $\mathbf{y} = \mathbf{y}(\mathbf{x})$ and $D_x \mathbf{F}, D_{\mathbf{y}} \mathbf{F}$, the Jacobian matrix of $\mathbf{y}$ is also continuous, therefore $\mathbf{y} = \mathbf{y}(\mathbf{x})$ is continuously differentiable.

Remark 5.11.

1. If F has higher order regularities, so does the implicit function.
2. One of main points of the Implicit Function Theorem which is very frequently used in mathematics is the following:

“Suppose $F(\mathbf{x}^0, \cdot)$ has a solution $\mathbf{y}^0$, then for $\mathbf{x}$ sufficiently close $\mathbf{x}^0$, $\mathbf{F}(\mathbf{x}, \cdot)$ also has a solution $\mathbf{y}$ close to $\mathbf{y}^0$. ”

This is often referred as “the parameter space supporting solutions is open!” If we can further prove this parameter space is closed, then we can prove that all parameter supports a solution.

One key in the proof of existence is to construct the following map:

$$\Phi_{\mathbf{x}} : \mathbf{y} \mapsto \mathbf{y} - (d_{\mathbf{y}} \mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)})^{-1} \mathbf{F}(\mathbf{x}, \mathbf{y})$$

depending on the parameter $\mathbf{x}$. The form of the map is very similarly to *Newton's Iteration* method:

$$y_{n+1} = y_n - f'(y_n)^{-1} f(y_n), \quad (5.1)$$

or more close to the *modified Newton's Iteration* method:

$$y_{n+1} = y_n - f'(y_0)^{-1} f(y_n). \quad (5.2)$$

Intuitively, since $d_{\mathbf{y}} \Phi_{\mathbf{x}} = \text{Id} - (d_{\mathbf{y}} \mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)})^{-1} \circ d_{\mathbf{y}} \mathbf{F}|_{(\mathbf{x}, \mathbf{y})}$ has operator norm small than $\frac{1}{2}$ for $(\mathbf{x}, \mathbf{y})$ sufficiently closed to $(\mathbf{x}^0, \mathbf{y}^0)$, $\Phi_{\mathbf{x}}$ is thus a contracting map by the *Mean Value Inequality*. Moreover, since $\Phi_{\mathbf{x}^0}(\mathbf{y}^0) = \mathbf{y}^0$, $\Phi_{\mathbf{x}^0}$ maps $B(\mathbf{y}^0, \delta)$ into $B(\mathbf{y}^0, \frac{\delta}{2})$. By the continuity of the map $\mathbf{F}(\mathbf{x}, \mathbf{y})$, $\Phi_{\mathbf{x}}$ maps $B(\mathbf{y}^0, \delta)$ to $B(\mathbf{y}^0, \frac{3}{4}\delta) \subset B(\mathbf{y}^0, \delta)$. This implies the existence of fixed point of $\Phi_{\mathbf{x}}$ by the Contracting Mapping Theorem, and the fixed point is precisely the solution to $\mathbf{F}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$.

Comparison between Implicit Function Theorem and Newton's Iteration Method, both providing a way of obtaining solutions to an equation:

- 牛顿迭代法属于无中生有；
- 隐函数定理由关于一个参数对应的方程的精确解得到附近参数对应的方程的“近似解”，然后再在此近似解附近找到此参数对应的方程的精确解；

- 有时会直接构造出关于某个参数所对应的方程的“近似解”，并证明导数的一致下界，这样的话也可以在此近似解附近找到精确解。这类做法更像是牛顿迭代法的过程。

3. The 3^{rd} condition means: the differential $d\mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)} : \mathbf{R}^n \times \mathbf{R}^m \rightarrow \mathbf{R}^m$ has full rank, i.e. $\text{rank } d\mathbf{F}|_{(\mathbf{x}^0, \mathbf{y}^0)} = m$.

If $d\mathbf{F}|_{M_0=(x_1^0, \dots, x_{n+m}^0)} : \mathbf{R}^{n+m} \rightarrow \mathbf{R}^m$ has full rank, then Linear Algebra tells us that there exists $1 \leq i_1 < \dots < i_m \leq n+m$ such that the corresponding partial differential $d_{(x_{i_1}, \dots, x_{i_m})}\mathbf{F}|_{M_0} : \mathbf{R}^m \rightarrow \mathbf{R}^m$ is invertible. We can use the above stated IFT to conclude that locally near M_0 , $\mathbf{F}(\mathbf{x}) = 0$ determines a vector-valued implicit function for which $\{x_1, \dots, x_{n+m}\} \setminus \{x_{i_1}, \dots, x_{i_m}\}$ are the independent variables and $\{x_{i_1}, \dots, x_{i_m}\}$ are the dependent variables.

4. By linear algebra, $\dim \ker d\mathbf{F}|_M = n+m - \text{rank } d\mathbf{F}|_M = n$, and the set $Z(\mathbf{F}) := \{\mathbf{x} \in \mathbf{R}^{n+m} | \mathbf{F}(\mathbf{x}) = \mathbf{0}\}$ is a n -dimensional object (as the graph of a n -variable vector-valued function). It is not difficult to see that

$$T_M(Z(\mathbf{F})) = \ker d\mathbf{F}|_M = \text{Null}(J_M(\mathbf{F})) = (\text{Col}(J_M(\mathbf{F})))^\perp = N_M(Z(\mathbf{F})),$$

in which $T_M(Z(\mathbf{F}))$ represents the tangent space of $Z(\mathbf{F})$ at M , and $N_M(Z(\mathbf{F}))$ represents the normal space to $Z(\mathbf{F})$ at M .

5.3 Several lower dimensional cases

It is easy to memorize how to calculate the differential of the implicit function. Plugging the implicit function to $\mathbf{F}(\mathbf{x}, \mathbf{y})$ we get

$$\mathbf{F}(\mathbf{x}, \mathbf{y}(\mathbf{x})) = \mathbf{0},$$

and differentiation with respect to $\mathbf{x}$ gives that

$$J_{\mathbf{x}}(\mathbf{F})d\mathbf{x} + J_{\mathbf{y}}(\mathbf{F})d\mathbf{y} = \mathbf{0},$$

thus

$$d\mathbf{y} = -(J_{\mathbf{y}}(\mathbf{F}))^{-1} J_{\mathbf{x}}(\mathbf{F})d\mathbf{x}.$$

- (I). $F : \mathbf{R}^3 \rightarrow \mathbf{R}$ is C^1 . If $F(x_0, y_0, z_0) = 0$ and $F'_z(x_0, y_0, z_0) \neq 0$, then there exists a neighborhood I of (x_0, y_0) in $\mathbf{R}^2$ and a neighborhood J of z_0 in $\mathbf{R}$ such that

$$\{F(x, y, z) = 0\} \cap (I \times J) = \{(x, y, z(x, y)) | (x, y) \in I\}.$$

And by differentiate $F(x, y, z(x, y)) = 0$ with respect to x and y we get

$$F'_x dx + F'_y dy + F'_z dz = 0,$$

i.e.

$$dz = -\frac{F'_x}{F'_z}dx - \frac{F'_y}{F'_z}dy.$$

Or, $\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}$ and $\frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$. This means that locally it is a differentiable surface. By item 4 of last section, we know that the normal direction to the surface at (x_0, y_0, z_0) is spanned by $\nabla F(x_0, y_0, z_0)$.

Example 5.12. Consider $F(x, y, z) = x^2 + y^2 - z^2$. At $(0, 0, 0)$, $\nabla F = 0$, we cannot use Implicit Function Theorem. We know that near the cone point, the surface cannot be represented as the graph of some function.

- (II). Let $F, G : \mathbf{R}^3 \rightarrow \mathbf{R}$ be two C^1 functions, consider $Z(F, G) = \{F = 0, G = 0\}$. Take the differential of $F(x, y, z) = 0, G(x, y, z) = 0$ then

$$\begin{pmatrix} F'_x & F'_y & F'_z \\ G'_x & G'_y & G'_z \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If $\begin{vmatrix} F'_y & F'_z \\ G'_y & G'_z \end{vmatrix} (x_0, y_0, z_0) \neq 0$, then locally we can solve an implicit function $x \mapsto (y, z) = (y(x), z(x))$ with

$$\begin{pmatrix} dy \\ dz \end{pmatrix} = - \begin{pmatrix} F'_y & F'_z \\ G'_y & G'_z \end{pmatrix}^{-1} \begin{pmatrix} F'_x \\ G'_x \end{pmatrix} (dx),$$

i.e.

$$\begin{cases} \frac{dy}{dx} = -\frac{F'_x G'_z - F'_z G'_x}{F'_y G'_z - F'_z G'_y} = -\frac{\frac{\partial(F, G)}{\partial(x, z)}}{\frac{\partial(F, G)}{\partial(y, z)}} \\ \frac{dz}{dx} = -\frac{F'_y G'_x - F'_x G'_y}{F'_y G'_z - F'_z G'_y} = -\frac{\frac{\partial(F, G)}{\partial(y, x)}}{\frac{\partial(F, G)}{\partial(y, z)}} \end{cases}$$

The set $Z(F, G)$ is a curve near (x_0, y_0, z_0) which can be parameterized by $(x, y(x), z(x))$. Tangent line has direction $\left(\frac{\partial(F, G)}{\partial(y, z)}, \frac{\partial(F, G)}{\partial(z, x)}, \frac{\partial(F, G)}{\partial(x, y)} \right) = \nabla F \times \nabla G$. Geometrically, this means that at a point $M \in Z(F) \cap Z(G)$, if $\nabla F \times \nabla G \neq \mathbf{0}$, i.e. the two surface $\{F = 0\}$ and $\{G = 0\}$ intersects “transversally” at M , then the intersection locus must be a differentiable curve.

- (III). Consider the map $(F, G) : \mathbf{R}^4 \rightarrow \mathbf{R}^2$ sending (x, y, u, v) to (F, G) which is assumed to be C^1 . If $\frac{\partial(F, G)}{\partial(u, v)}(M) \neq 0$, the equations

$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$$

will define an implicit function $(x, y) \mapsto (u, v) = (u(x, y), v(x, y))$ whose graph in $\mathbf{R}^4$ is a two dimensional surface. By solving

$$\begin{pmatrix} F'_x & F'_y & F'_u & F'_v \\ G'_x & G'_y & G'_u & G'_v \end{pmatrix} \begin{pmatrix} dx \\ dy \\ du \\ dv \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

we obtain that

$$\begin{aligned} \begin{pmatrix} du \\ dv \end{pmatrix} &= - \begin{pmatrix} F'_u & F'_v \\ G'_u & G'_v \end{pmatrix}^{-1} \begin{pmatrix} F'_x & F'_y \\ G'_x & G'_y \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} \\ &= - \frac{1}{\frac{\partial(F,G)}{\partial(u,v)}} \begin{pmatrix} \frac{\partial(F,G)}{\partial(x,v)} & \frac{\partial(F,G)}{\partial(y,v)} \\ \frac{\partial(F,G)}{\partial(x,u)} & \frac{\partial(F,G)}{\partial(y,u)} \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} \end{aligned} \quad (5.3)$$

Example 5.13. Let z be the implicit function determined by $e^z - xyz = 0$. Find $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

Example 5.14. Let z be the implicit function determined by $ax^2 + by^2 + cz^2 = 1$. Find $\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y^2}$.

Example 5.15. The equations $u^2 - v + x = 0, u + v^2 - y = 0$ determines an implicit function $(x, y) \mapsto (u, v)$ near $(0, 0)$. Calculate $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$.

5.4 Inverse Function Theorem

Theorem 5.16 (Inverse Function Theorem). *Let $\mathbf{F} : D(\subset \mathbf{R}^n) \rightarrow \mathbf{R}^n$ be a C^1 map, $\mathbf{F}(\mathbf{x}^0) = \mathbf{y}^0$. Suppose $d\mathbf{F}|_{\mathbf{x}^0} : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is invertible, i.e. $J_{\mathbf{x}^0}(\mathbf{F})$ is invertible, then there exists an open neighborhood U of $\mathbf{x}^0$ and an open neighborhood V of $\mathbf{y}^0$ such that*

$$\mathbf{F}|_U : U \rightarrow V$$

is one-to-one, and the inverse

$$\mathbf{G} := (\mathbf{F}|_U)^{-1} : V \rightarrow U$$

is C^1 with

$$d\mathbf{G}|_{\mathbf{y}} = (d\mathbf{F}|_{\mathbf{x}=\mathbf{G}(\mathbf{y})})^{-1}.$$

Proof. Define a map $\mathbf{H} : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}^n$ sending $(\mathbf{x}, \mathbf{y})$ to $\mathbf{y} - \mathbf{F}(\mathbf{x})$. We then just need to realize that

$$d_{\mathbf{x}}\mathbf{H}|_{(\mathbf{x}, \mathbf{y})} = -d\mathbf{F}|_{\mathbf{x}}$$

is invertible at $(\mathbf{x}^0, \mathbf{y}^0)$. Since $\mathbf{H}(\mathbf{x}^0, \mathbf{y}^0) = \mathbf{0}$ and $\mathbf{H}$ is C^1 , we can use Implicit Function Theorem to conclude that there exists an open neighborhood U' of $\mathbf{x}^0$ and an open neighborhood V of $\mathbf{y}^0$ such that $\forall \mathbf{y} \in V, \exists! \mathbf{x} \in U'$ such that $\mathbf{H}(\mathbf{x}, \mathbf{y}) = \mathbf{0}$. Let us denote this implicit function by $\mathbf{G} : V \rightarrow U'$, and

$U := \mathbf{G}(V)$.

Next we prove that U is open. For $\mathbf{x}' \in U$ saying $\mathbf{x}' = \mathbf{G}(\mathbf{y}')$ with $\mathbf{y}' \in V$. Since V is open, there exists $\varepsilon > 0$ such that $B(\mathbf{y}', \varepsilon) \subset V$. By the continuity of $\mathbf{F}$, there exists $\delta > 0$ such that $\mathbf{F}(B(\mathbf{x}', \delta)) \subset B(\mathbf{y}', \varepsilon) \subset V$ and $B(\mathbf{x}', \delta) \subset U'$.

For any $\mathbf{x}'' \in B(\mathbf{x}', \delta)$, $\mathbf{F}(\mathbf{x}'') \in V$ and $\mathbf{F}(\mathbf{G}(\mathbf{F}(\mathbf{x}''))) = \mathbf{F}(\mathbf{x}'')$. By the uniqueness of $\mathbf{x} \in U'$ such that $\mathbf{H}(\mathbf{x}, \mathbf{F}(\mathbf{x}'')) = 0$, we get that $\mathbf{x}'' = \mathbf{G}(\mathbf{F}(\mathbf{x}''))$. This implies that $\mathbf{x}'' \in \mathbf{G}(V) = U$ is open. In conclusion, $\mathbf{F} : U \rightarrow V$ and $\mathbf{G} : V \rightarrow U$ are inverse to each other, and

$$d\mathbf{G}|_{\mathbf{y}} = -(d_{\mathbf{x}}\mathbf{H})^{-1} \circ d_{\mathbf{y}}\mathbf{H}|_{(\mathbf{G}(\mathbf{y}), \mathbf{y})} = (d\mathbf{F}|_{\mathbf{x}=\mathbf{G}(\mathbf{y})})^{-1}.$$

□

Remark 5.17. Actually, Inverse Function Theorem can be used to prove Implicit Function Theorem: construct the function $\mathbf{H}(\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \mathbf{F}(\mathbf{x}, \mathbf{y}))$. The inverse function restricted to open neighborhood of the form $U \times V$ gives the *straightening* of the level sets $\{F^{-1}(\mathbf{z}) | \mathbf{z} \in V\}$ (restricted to $\mathbf{x} \in U$).

Example 5.18 (Coordinate transformation). Consider the map $(r, \theta) \mapsto (x, y) = (r \cos \theta, r \sin \theta)$. The Jacobian matrix is

$$\begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

so the Jacobian determinant is r , thus the map is locally invertible with a C^1 inverse. (Pictorial the map $(0, +\infty) \times \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}$).